Student Performance Analytics System

Python with AI Project Documentation
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Batch: Batch 2
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1. Objective

To analyse student marks and attendance using Python, NumPy and Pandas, and report totals, averages, grades, pass/fail status, subject performance and top performers.

2. Technologies Used

Python is used for program flow, conditions, loops and reusable functions. Pandas reads and organises CSV records in DataFrames, groups records and exports results. NumPy performs numerical calculations such as mean, maximum and minimum.

3. Dataset Description

The project includes 20 fictional sample student records. Each record contains Student ID, Name, Department, marks in Python, Mathematics, Database and AI, and Attendance Percentage. These are demonstration records, not real institutional data.

4. Implementation

First, main.py loads students.csv into a Pandas DataFrame. The validate_dataset() function checks required columns and ensures marks and attendance are between 0 and 100. The process_students() function calculates total and average marks, assigns grades and determines pass/fail status. The get_subject_summary() function uses a loop and NumPy to calculate subject averages, highest marks and lowest marks. The get_class_summary() function calculates class average, highest and lowest student averages, pass/fail counts and top performers. Pandas groupby() calculates department averages. Results are printed and exported to CSV files.

5. Grading and Pass/Fail Rules

Grades: A for average >= 85; B for >= 70; C for >= 55; D for >= 40; otherwise F. A student passes only if every subject mark is at least 40 and attendance is at least 75%. These are the rules implemented in this version; confirm with the trainer if a different rule was specified in class.

6. Key Features

CSV dataset loading and validation; total and average marks; grade assignment; pass/fail analysis; class and subject statistics; department-wise averages; top performer identification; CSV export of results.

7. Output / Screenshots

```
PS D:\Documents\PythonAI\student-performance-analytics> python main.py
======================================================================
           STUDENT PERFORMANCE ANALYTICS SYSTEM
======================================================================

Students analysed: 20

STUDENT RESULTS
Student_ID          Name             Department  Total_Marks  Average_Marks Grade Result
      S001   Aarav Sharma   Computer Engineering          343          85.75     A   Pass
      S002   Ananya Patel Information Technology          322          80.50     B   Pass
      S003 Rohan Deshmukh   Computer Engineering          253          63.25     C   Pass
      S004  Isha Kulkarni            Electronics          368          92.00     A   Pass
      S005   Vivaan Joshi Information Technology          205          51.25     D   Fail
      S006     Meera Shah   Computer Engineering          331          82.75     B   Pass
      S007    Aditya More             Mechanical          169          42.25     D   Fail
      S008      Sara Khan            Electronics          312          78.00     B   Pass
      S009    Arjun Patil             Mechanical          270          67.50     C   Pass
      S010      Diya Nair Information Technology          368          92.00     A   Pass
      S011    Kabir Singh   Computer Engineering          204          51.00     D   Fail
      S012      Aditi Rao            Electronics          341          85.25     A   Pass
      S013  Reyansh Gupta             Mechanical          278          69.50     C   Pass
      S014     Myra Joshi Information Technology          260          65.00     C   Pass
      S015   Ishaan Verma   Computer Engineering          371          92.75     A   Pass
      S016    Kiara Mehta            Electronics          218          54.50     D   Fail
      S017   Atharv Pawar             Mechanical          144          36.00     F   Fail
      S018     Navya Iyer Information Technology          332          83.00     B   Pass
      S019     Dhruv Sane   Computer Engineering          276          69.00     C   Pass
      S020  Tara Banerjee            Electronics          358          89.50     A   Pass

CLASS SUMMARY
Class average:       71.54/100
Highest average:     92.75/100
Lowest average:      36.00/100
Students passed:     15
Students failed:     5
Top performer(s):   Ishaan Verma

SUBJECT-WISE SUMMARY
    Subject  Average  Highest  Lowest
     Python    68.90       95      28
Mathematics    71.20       94      38
   Database    70.55       95      33
         AI    75.50       96      45

DEPARTMENT-WISE AVERAGE
Department
Electronics               79.85
Information Technology    74.35
Computer Engineering      74.08
Mechanical                53.81

OUTPUT FILES
Detailed results: D:\Documents\PythonAI\student-performance-analytics\output\student_results.csv
Subject summary:  D:\Documents\PythonAI\student-performance-analytics\output\subject_summary.csv

Analysis completed successfully.
PS D:\Documents\PythonAI\student-performance-analytics>
```

8. Final Outcome

The program successfully processes the sample dataset and produces student-level results, class statistics, subject comparisons and department averages. It demonstrates Python fundamentals, NumPy and Pandas in a practical data-analysis task.

9. Challenges and Learning

The project demonstrates the importance of consistent CSV structure, validating values before calculations, separating logic into reusable functions, using Pandas for tabular analysis and NumPy for numerical calculations, and exporting results for review. Edit this section to include the actual challenges you personally encountered.

10. GitHub Repository

Repository URL: https://github.com/Pradnya1227/student-performance-analytics